Lab 02 Report: Comparing Sentiment Polarity and Noun Density Across Two Blogs

Name: Hadia Imran
SAP ID: 70177704
Course: Introduction to AI (AI-101L)
Date: 1 November 2026

---

1. Introduction

Most real-world text data is unstructured and must be collected and converted into numbers before it can be analysed. This assignment applies the full Lab 02 pipeline: scraping articles from the web, engineering numerical features with NLP tools, and exploring the results visually.

Two blogs are compared on two measures:

1. **Sentiment polarity**: how positive or negative the writing is (-1 to +1).
2. **Noun density**: the share of tokens that are nouns, used as a rough indicator of how informational or descriptive the writing is.

Research question: Do the two blogs differ in tone (polarity) and in how noun-heavy their writing is?

2. Data Collection

- **Blog A:** TechnCruncher (techncruncher.blogspot.com), 10 articles.
- **Blog B:** BloggerBuzz (blogger.googleblog.com), 10 articles.
- **Total:** 20 articles.

Both are Blogger-hosted, so articles were collected through Blogger's JSON feed (`/feeds/posts/default`) with the `requests` library. The HTML body of each post was parsed with BeautifulSoup and the text of all paragraph tags was joined into one string per article. Posts shorter than 300 characters were discarded as quality control, and a one-second delay was used between requests. The final dataset (`url`, `blog`, `title`, `text` and all engineered features) is saved as `articles.csv`.

3. Feature Engineering

Feature	Tool	Definition
`num_tokens`	spaCy	Words and punctuation marks
`num_sentences`	spaCy	Sentences found by sentence segmentation
`num_entities`	spaCy	Named entities (people, organisations, places)
`num_nouns`	spaCy	Tokens tagged NOUN or PROPN
`noun_density`	derived	`num_nouns / num_tokens`
`sentiment_polarity`	TextBlob	-1 (negative) to +1 (positive)
`sentiment_subjectivity`	TextBlob	0 (objective) to 1 (subjective)
`title_length`	pandas	Characters in the title

Noun density is used instead of the raw noun count because articles differ greatly in length, so longer articles would otherwise always appear to have more nouns.

4. Exploratory Data Analysis and Results

4.1 Summary statistics

> Values below are read from Figure 4 and are approximate. Replace them with the exact numbers printed by the notebook's summary cell.

Blog	Mean polarity	Std polarity	Mean noun density	Std noun density	Typical length (tokens)
A: TechnCruncher	≈ 0.18	≈ 0.05	≈ 0.29	≈ 0.02	≈ 1,000 to 1,800 (one outlier ≈ 9,000)
B: BloggerBuzz	≈ 0.22	≈ 0.14	≈ 0.25	≈ 0.05	≈ 200 to 500

Welch's t-test (replace with the notebook output): polarity t = [ ], p = [ ] (approx. p ≈ 0.4); noun density t = [ ], p = [ ] (approx. p ≈ 0.04).

4.2 Polarity distribution (Figure 1)

Figure 1: Sentiment polarity distribution by blog

Both blogs are overwhelmingly positive in tone: almost every article scores above zero. TechnCruncher's scores are tightly grouped, mostly between about 0.1 and 0.25. BloggerBuzz is

much more spread out, running from roughly 0 to about 0.42, and includes one article with a slightly negative score. Its distribution sits a little to the right of TechnCruncher's, but the two overlap heavily.

4.3 Pair plot of derived features (Figure 2)

Figure 2: Pair plot of derived features

The pair plot shows several patterns:

- **Length:** TechnCruncher articles are much longer (mostly 1,000 to 1,800 tokens, with one very long post of about 9,000 tokens and about 350 entities). BloggerBuzz articles are short (roughly 200 to 500 tokens).
- **Tokens and entities:** longer articles contain more named entities, which is expected, and the single long TechnCruncher post stretches this trend.
- **Noun density:** TechnCruncher forms a tight cluster (about 0.26 to 0.32). BloggerBuzz is scattered, and three of its posts have unusually low noun density (about 0.17 to 0.18).
- **Noun density vs polarity:** those three low-noun-density BloggerBuzz posts are also among its more positive ones, which hints that more conversational writing reads as more positive. With so few points this is only a tentative observation.
- **Titles:** BloggerBuzz titles tend to be shorter than TechnCruncher's.

4.4 TF-IDF top terms (Figure 3)

Figure 3: Top 20 TF-IDF terms

The highest-scoring terms are content, google, ai and new, followed by make, help, use, tools and tool. Terms such as creators, affiliate, cj, features and platform point to the marketing and AI-tools topics of TechnCruncher, while google, search, data, models and information point to the Google-product and technology announcements typical of BloggerBuzz. Overall the corpus is dominated by technology and content-creation vocabulary.

4.5 Blog comparison (Figure 4)

Figure 4: Mean polarity and noun density by blog

Figure 4 compares the blogs side by side on both metrics, with error bars showing one standard deviation.

- **Polarity:** BloggerBuzz has a slightly higher mean (about 0.22 vs 0.18), but its error bar is nearly three times wider. The difference is small relative to the spread and is **not statistically significant** ($p \approx 0.4$).
- **Noun density:** TechnCruncher is higher (about 0.29 vs 0.25) and far more consistent. This difference is larger relative to the spread and is **likely significant** at the 5% level ($p \approx 0.04$).

5. Discussion

Noun density is the clearer difference between the two blogs. TechnCruncher's writing is consistently noun-heavy, which fits its "top 10 AI tools" and review format: lists of product names, features and platforms. BloggerBuzz has lower and more variable noun density, which suggests a

more mixed style that includes conversational and announcement-style posts with more verbs, pronouns and adjectives.

For sentiment, both blogs are positive, as expected for promotional and product-announcement writing. BloggerBuzz's slightly higher average is not reliable, because a few very positive posts and one slightly negative post produce a wide spread. The more useful finding is that TechnCruncher's tone is very uniform, which is typical of formulaic review articles.

An important confound is **article length**. BloggerBuzz posts are several times shorter than TechnCruncher posts, and TextBlob and noun-density estimates are noisier on short texts. Part of BloggerBuzz's larger variance is therefore likely due to length rather than genuine differences in style.

6. Limitations

- **Small sample:** 10 articles per blog is too few for strong conclusions, and the t-tests have low power.
- **Unequal article length:** the blogs differ greatly in post length, and one TechnCruncher post (about 9,000 tokens) is a clear outlier.
- **Lexicon-based sentiment:** TextBlob scores words from a dictionary, so it misses sarcasm, negation and context, and promotional wording inflates positivity.
- **POS tagging errors:** spaCy's small model (`en_core_web_sm`) is fast but imperfect, especially on brand names and informal text.
- **Non-random sample:** only the most recent posts from each feed were collected.
- **Boilerplate text:** calls to action or leftover template text can influence both metrics.

7. Conclusion

Across 20 articles, the two blogs show similar, positive sentiment (mean polarity about 0.18 for TechnCruncher and 0.22 for BloggerBuzz, not a significant difference), but TechnCruncher uses a noticeably higher share of nouns (about 0.29 vs 0.25), consistent with its product-focused listicle style. BloggerBuzz is more varied on both measures, partly because its posts are much shorter. Given the small sample and uneven article lengths, these findings are indicative rather than conclusive, and a larger and length-matched corpus would be needed to confirm them.

8. Files Submitted

- `scraping_eda.ipynb`: full code
- `articles.csv`: scraped data and features
- `fig1_polarity_distribution.png`, `fig2_pairplot.png`, `fig3_tfidf_top_terms.png`, `fig4_grouped_bar_comparison.png`
- `report.md`: this report